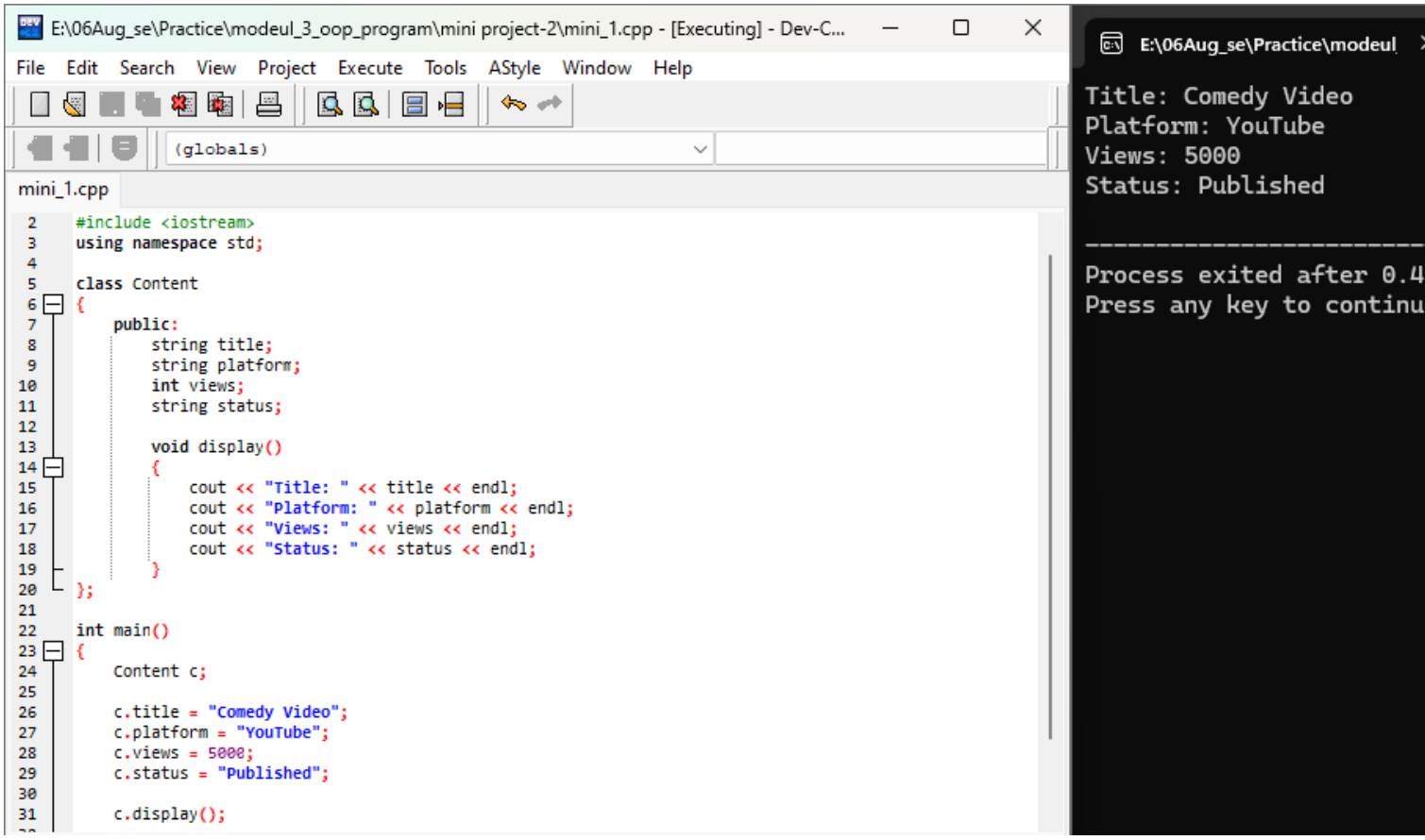
MINI PROJECT -2

Q 1. Create a Content class with properties: title, platform, views, and status. Write a method to display all details of a Content object.

Ans.




```
#include <iostream>
using namespace std;
 class Content
{
```

```
    public:
        string title;
        string platform;
        int views;
        string status;
        void display()
        {
            cout << "Title: " << title << endl;
            cout << "Platform: " << platform << endl;
            cout << "Views: " << views << endl;
            cout << "Status: " << status << endl;
        }
};
int main()
{
    Content c;
    c.title = "Comedy Video";
    c.platform = "YouTube";
    c.views = 5000;
    c.status = "Published";

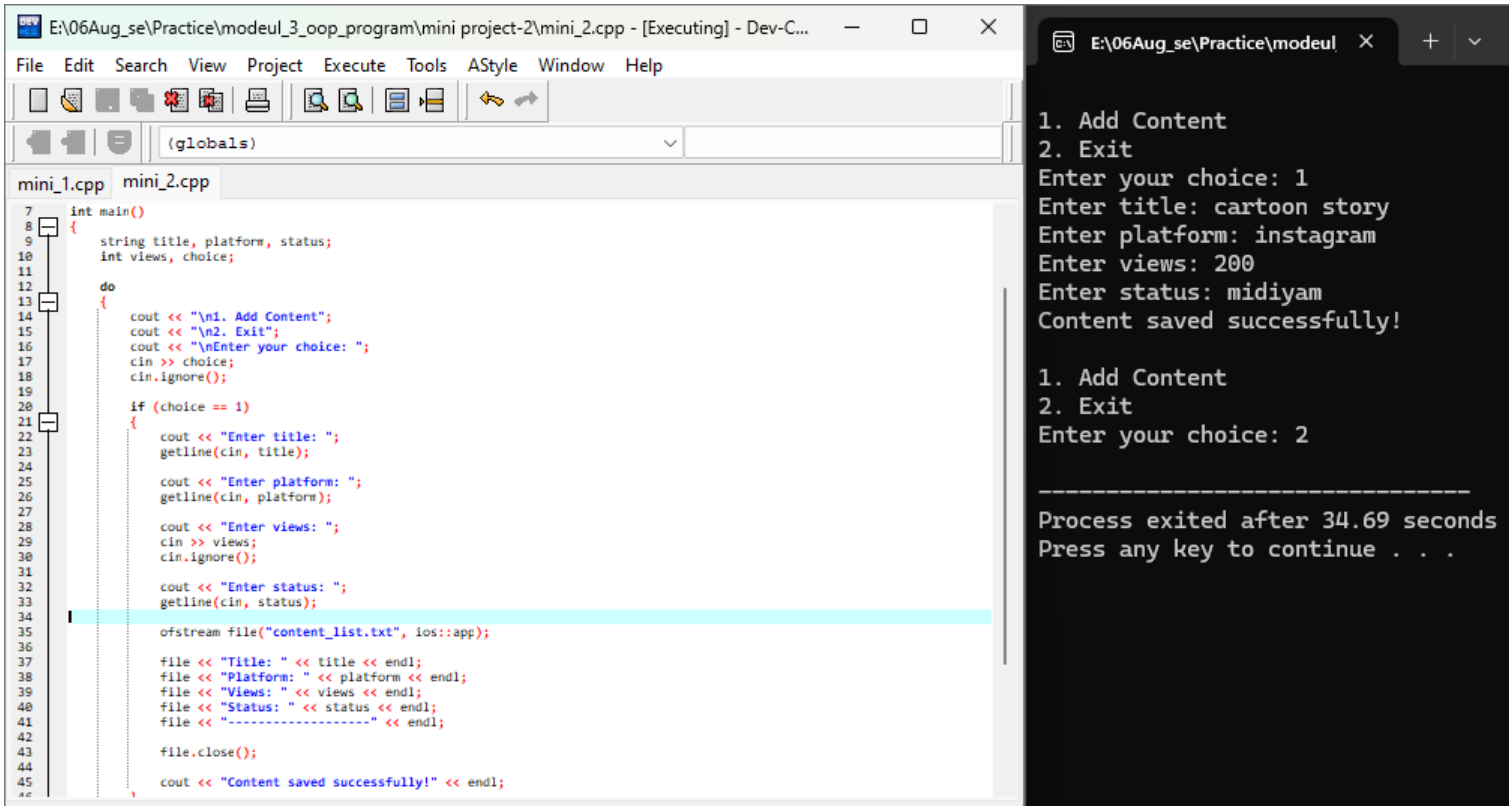
    c.display();

    return 0;
}
```

Q 2. Build a console menu that lets users add new content ideas (title, platform, views, status) and save each entry to a text file named content_list.txt.

Hint: Use file handling to append each new content item to the file..

Ans.




```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
int main()
```

```
{
    string title, platform, status;
    int views, choice;
     do
    {
        cout << "\n1. Add Content";
        cout << "\n2. Exit";
        cout << "\nEnter your choice: ";
        cin >> choice;
        cin.ignore();

        if (choice == 1)
        {
            cout << "Enter title: ";
            getline(cin, title);

            cout << "Enter platform: ";
            getline(cin, platform);

            cout << "Enter views: ";
            cin >> views;
            cin.ignore();

            cout << "Enter status: ";
            getline(cin, status);
```

```
            ofstream file("content_list.txt", ios::app);

            file << "Title: " << title << endl;
            file << "Platform: " << platform << endl;
            file << "Views: " << views << endl;
            file << "Status: " << status << endl;
            file << "------------------" << endl;

            file.close();

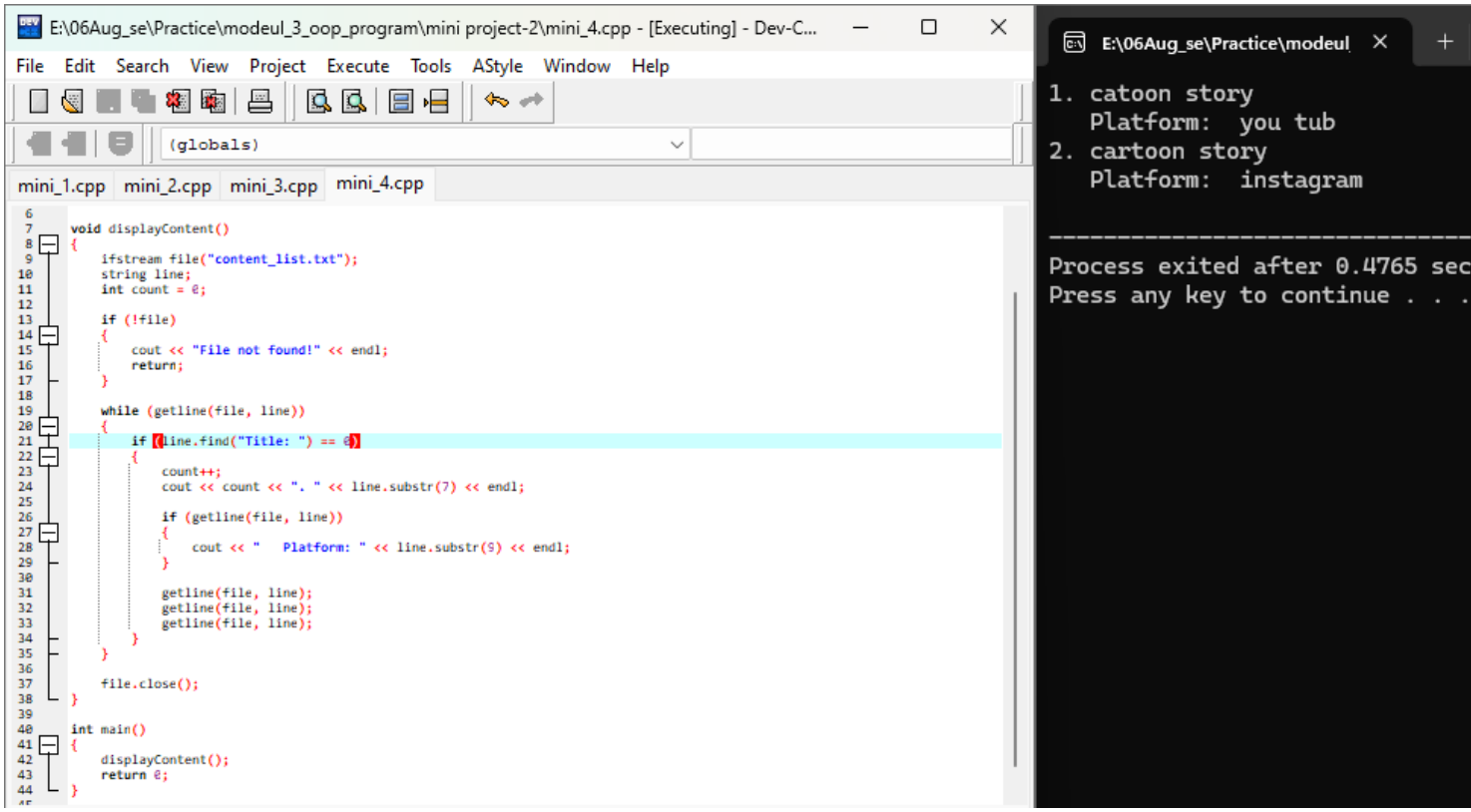
            cout << "Content saved successfully!" << endl;
        }
        else if (choice != 2)
        {
            cout << "Invalid choice!" << endl;
        }

    } while (choice != 2);

    return 0;
}
```

Q 3. Implement a function to read all content items from content_list.txt and display them in a numbered list, showing title and platform for each.

Ans.




```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

void displayContent()
{
    ifstream file("content_list.txt");
```

```
string line;
int count = 0;

if (!file)
{
    cout << "File not found!" << endl;
    return;
}

while (getline(file, line))
{
    if (line.find("Title: ") == 0)
    {
        count++;
        cout << count << ". " << line.substr(7) << endl;

        if (getline(file, line))
        {
            cout << "   Platform: " << line.substr(9) << endl;
        }

        getline(file, line);
        getline(file, line);
        getline(file, line);
    }
```

```
    }

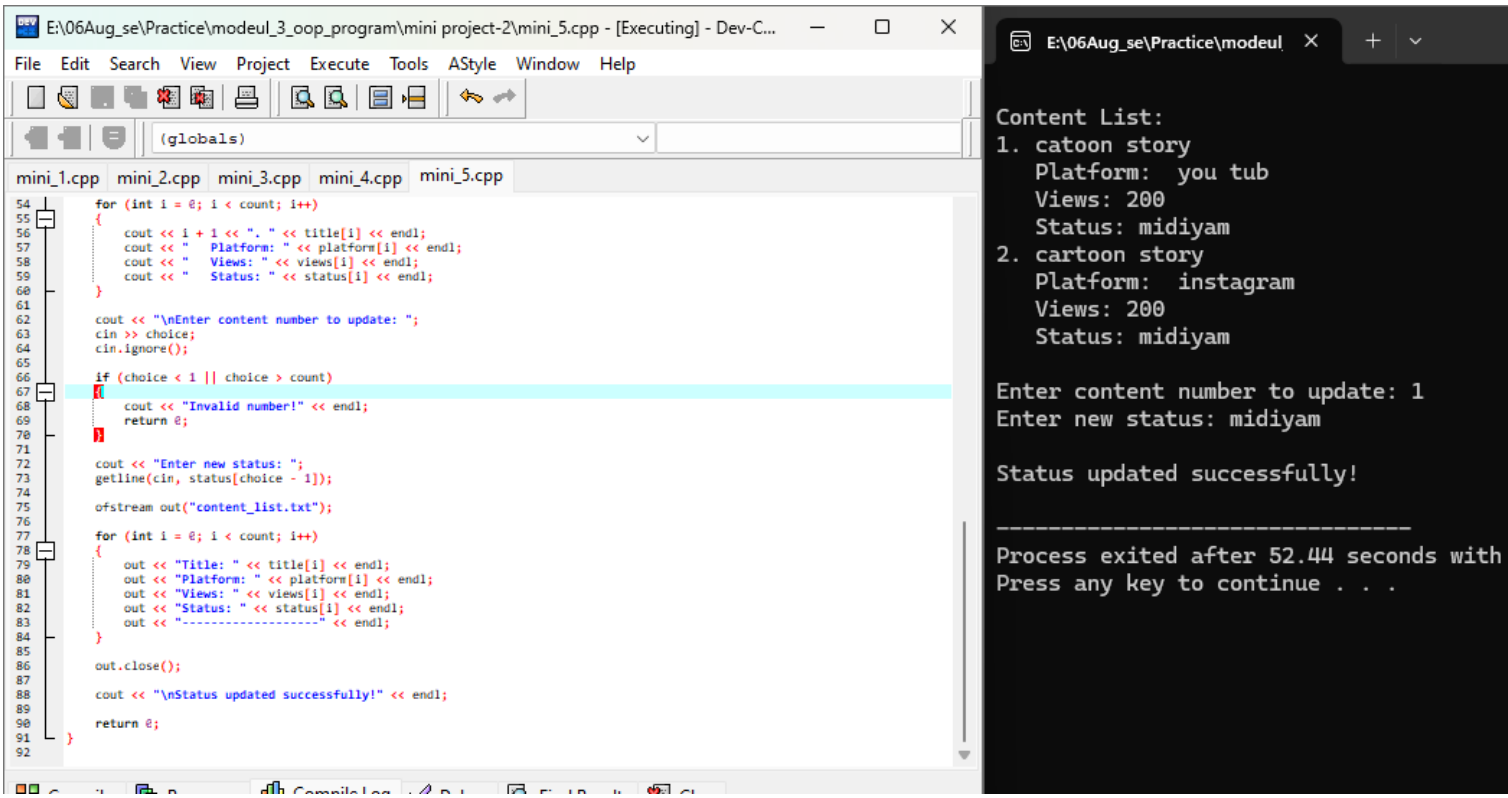
    file.close();
}

int main()
{
    displayContent();
    return 0;
}
```

Q 4. Add an option in your console app to let users update the status of any content idea by selecting its number from the displayed list and saving the change back to the file.

Hint: You will need to read all items, modify the selected one, and overwrite the file..

Ans.




```
#include <iostream>
#include <fstream>
#include <string>
#include <cstdlib>
```

```
using namespace std;

int main()
{
    string title[100], platform[100], status[100];
    int views[100];
    int count = 0, choice;
    string line;

    ifstream file("content_list.txt");

    if (!file)
    {
        cout << "File not found!" << endl;
        return 0;
    }

    while (count < 100 && getline(file, line))
    {
        if (line.find("Title: ") == 0)
        {
            title[count] = line.substr(7);
```

```
            getline(file, line);
            platform[count] = line.substr(9);

            getline(file, line);
            views[count] = atoi(line.substr(7).c_str());

            getline(file, line);
            status[count] = line.substr(8);

            getline(file, line); // Separator

            count++;
        }
    }

    file.close();

    if (count == 0)
    {
        cout << "No content found!" << endl;
        return 0;
```

```
}

cout << "\nContent List:\n";

for (int i = 0; i < count; i++)
{
    cout << i + 1 << ". " << title[i] << endl;
    cout << "   Platform: " << platform[i] << endl;
    cout << "   Views: " << views[i] << endl;
    cout << "   Status: " << status[i] << endl;
}

cout << "\nEnter content number to update: ";
cin >> choice;
cin.ignore();

if (choice < 1 || choice > count)
{
    cout << "Invalid number!" << endl;
    return 0;
}
```

```
    cout << "Enter new status: ";
    getline(cin, status[choice - 1]);

    ofstream out("content_list.txt");

    for (int i = 0; i < count; i++)
    {
        out << "Title: " << title[i] << endl;
        out << "Platform: " << platform[i] << endl;
        out << "Views: " << views[i] << endl;
        out << "Status: " << status[i] << endl;
        out << "-------------------" << endl;
    }

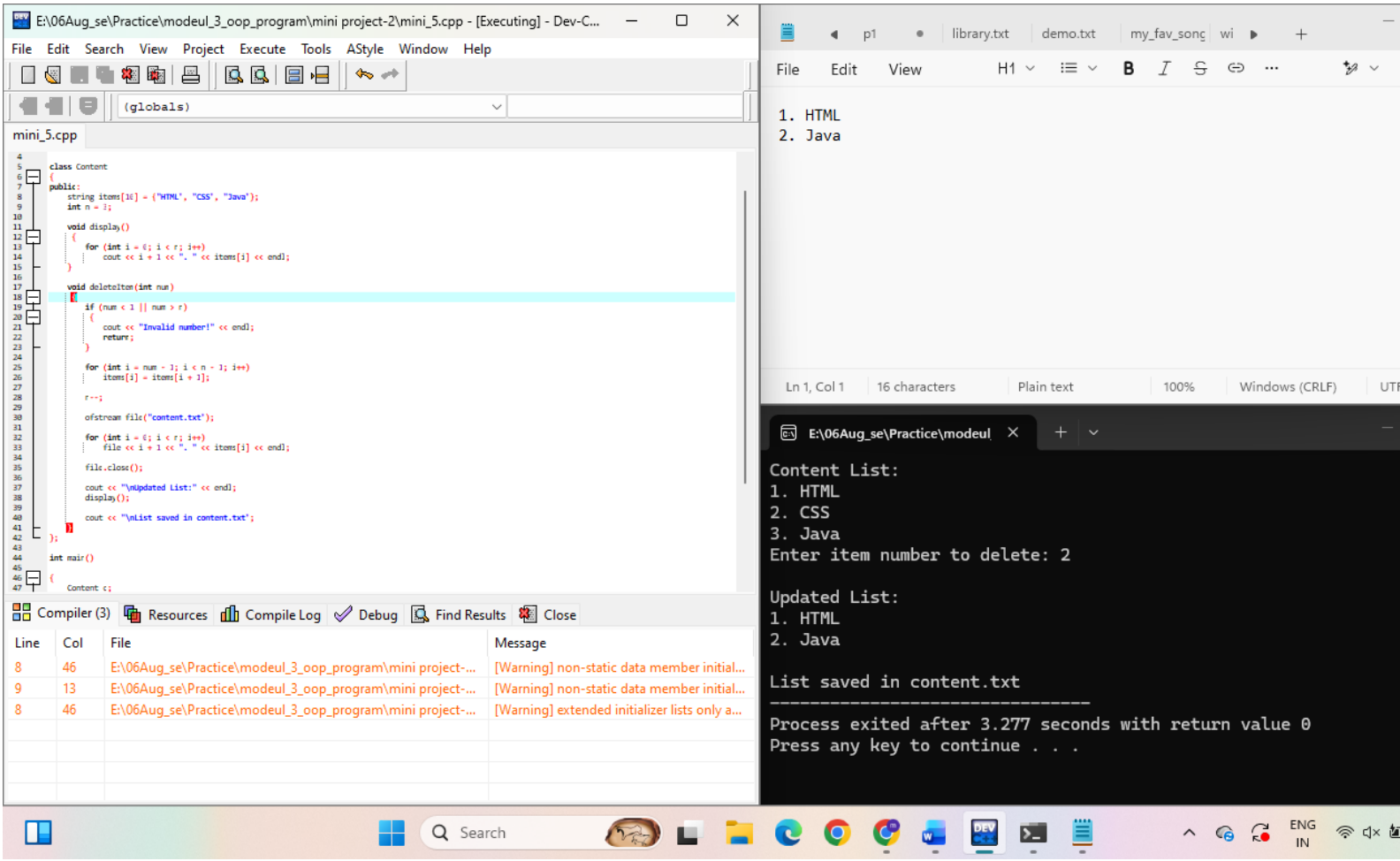
    out.close();

    cout << "\nStatus updated successfully!" << endl;

    return 0;
}
```

Q 5. Add a feature to delete a content item by its number from the list, update the file accordingly, and display the updated list to confirm deletion.

Ans.




```
#include <iostream>
#include <fstream>
using namespace std;

class Content
```

```
{
public:
    string items[10] = {"HTML", "CSS", "Java"};
    int n = 3;

    void display()
       {
        for (int i = 0; i < n; i++)
            cout << i + 1 << ". " << items[i] << endl;
    }

    void deleteItem(int num)
       {
        if (num < 1 || num > n)
             {
            cout << "Invalid number!" << endl;
            return;
        }

        for (int i = num - 1; i < n - 1; i++)
            items[i] = items[i + 1];
```

```
        n--;

        ofstream file("content.txt");

        for (int i = 0; i < n; i++)
            file << i + 1 << ". " << items[i] << endl;

        file.close();

        cout << "\nUpdated List:" << endl;
        display();

        cout << "\nList saved in content.txt";
    }
};

int main()

{
    Content c;
    int num;
```

```
    cout << "Content List:" << endl;
    c.display();

    cout << "Enter item number to delete: ";
    cin >> num;

    c.deleteItem(num);

    return 0;
}
```